

Subject Description Form

Subject Code	COMP5567
Subject Title	Distributed Algorithms and Protocols for Blockchains
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	The objectives of this subject are to: • introduce the principles, concepts, theories, and analysis of distributed algorithms for blockchain systems; • enable the students to analyze the underlying distributed consensus protocols of popular blockchain platforms; and • enable the students to develop basic distributed consensus protocols for blockchain
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) demonstrate a comprehensive understanding of distributed consensus algorithms, including the algorithm design, correctness proof, and complexity analysis; b) possess the ability to analyze the distributed consensus protocols in existing popular blockchain platforms, including Bitcoin, Ethereum, and Hyperledger Fabric; and c) possess the ability of developing basic distributed consensus protocols for blockchains using remote procedure call tools; and d) understand the advanced blockchain consensus protocols with high throughput and high scalability.
Subject Synopsis/ Indicative Syllabus	Introduction to Blockchain Consensus Architecture of a typical blockchain systems; basics of blockchain consensus; history of consensus protocols; history of blockchain consensus; relationship between traditional consensus and blockchain consensus. Distributed Consensus Algorithms System model and problem definition; properties; termination; agreement; integrity; faulty models; Byzantine general problem;

	interactive consistency; impossibility results; synchronous algorithms; correctness analysis; complexity analysis; case studies. Proof of Work, Proof of Stake, and Practical Byzantine Fault Tolerance Hash puzzles; difficulties; forking problem; classic proof of work; ghost protocol; analysis of proof of work protocols; classic proof of stake; centralization issue; improvement proposal; impossibility of byzantine fault tolerance; practical byzantine fault tolerance; comparison and demonstration. Bitcoin, Ethereum, and Public Blockchains Architecture of a Bitcoin node; architecture of an Ethereum node; bootstrapping; mining pool; selfish mining; analysis of Bitcoin source code; analysis of Ethereum source code. Hyperledger Fabric and Federated Blockchains Architecture of Hyperledger Fabric; peers; endorsers; orderer; ordering services; Kafka; zookeeper; practical byzantine fault tolerance; analysis of Hyperledger Fabric source code. High Performance Blockchain Consensus Throughput and scalability; payment channel networks (PCN); limitations of traditional consensus protocols; hybrid consensus; Bitcoin-NG; parallelization; sharding; trusted execution environments; DAG consensus. Algorithms and Protocols for Multiple Chains and Web 3.0 Cross-chain operations; consensus among multiple chains; distributed algorithms in Web 3.0; peer-to-peer network; protocol implementation; case studies.
Teaching/Learning Methodology	39 hours of class activities including lectures, tutorials, and labs. • The lectures will be used to deliver course materials; • The tutorials will be used to show case studies to consolidate the understanding of course materials; • The labs will be used to give hands-on guidance to practice exercises.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed			
			a	b	c	d
	1. Assignments, lab and project	30%	√	√	√	√
	2. Exam	70%	√	√		√
	Total	100 %				
	Assignments, lab and exam will assess students’ understanding of distributed consensus algorithms (a), blockchain consensus protocols (b), and high-performance blockchain consensus protocols (d). The project will be on a group basis to assess students’ sense of teamwork and abilities in implementing and developing distributed consensus protocols (c). Each group needs a single submission containing the necessary documents, source codes, etc., as requested, as well as the percentages of contributions for each member (signed by all the members).					
Student Study Effort Expected	Class contact:					
	• Class activities (lectures, tutorials, and labs)		39 Hrs.			
	Other student study effort:					
	• Assignments, Projects, Exams		66 Hrs.			
	Total student study effort		105 Hrs.			
Reading List and References	Books					
	Lynch, N. A. (1996). Distributed algorithms. Elsevier.					
	Bashir, I., 2020. Mastering Blockchain: A deep dive into distributed ledgers, consensus protocols, smart contracts, DApps, cryptocurrencies, Ethereum, and more. Packt Publishing Ltd.					
	Conference Proceedings and Journals:					
	ACM SIGSAC Conference on Computer and Communications Security (CCS)					
	ACM International Conference on Management of Data (SIGMOD)					

	IEEE Conference on Computer Communications (INFOCOM) IEEE Transactions on Computers IEEE Transactions on Parallel and Distributed Systems Online Materials: https://github.com/bitcoin/bitcoin https://github.com/ethereum/go-ethereum https://github.com/hyperledger/fabric
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